Problem:

sudo ip link set enp0s3 up

sudo nano /etc/netplan/01-network-manager-all.yaml

```
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    enp0s3:
      dhcp4: true
```

sudo netplan apply

sudo nano /etc/NetworkManager/NetworkManager.conf

sudo systemctl restart NetworkManager

sudo dhclient -v enp0s3

Exp 1:

```
sudo useradd -m Nanthitha
sudo passwd Nanthitha
sudo usermod -aG sudo Nanthitha
su - Nanthitha
sudo deluser --remove-home Nanthitha
pwd
ls
echo “hello” >> nan.txt
grep “hello” nan
df -h
du -sh
top
date
time
touch file.txt
head -n file.txt
uname -a
chmod 777 file.txt
ls -l file.txt
find . -name file.txt
locate file.txt
```

Exp2:

sudo apt install default-jdk -y

sudo apt install --reinstall java-wrappers

sudo apt install --reinstall default-jdk

Sudo ufw disable
Sudo iptables -F

Import java.util.Scanner
System.nanoTime()
System.out.printf("%.6f seconds\n", time);

C pgm
#include <time.h>
Clock_t start=clock()
(double)(et-st)/CLOCKS_PER_SEC

Gcc -o mypgm file.c
./mypgm

Ep 5: kvm

Egrep -c 'lm' /proc/cpuinfo
Uname -a
Ls /lib/modules/5.4.0-84-generic/kernel/arch/x86/kvm/kvm*
Sudo apt-get install qemu-kvm libvirt-bin bridge-utils qemu-sys
sudo apt-get install qemu-kvm libvirt-daemon-system libvirt-clients bridge-utils qemu-system-x86

Sudo nano /etc/libvirt/libvirtd.conf
Grep ^[^#] /etc/libvirt/libvirtd.conf

unix_sock_group="libvirt"
unix_sock_ro_perms="0777"
unix_sock_rm_perms="0770"
unix_sock_dir="/var/run/libvirt"
auth_unix_ro="none"

auth_unix_rw=”none”
Sudo systemctl restart libvirtd
Virsh list
Virsh
Version
Nodeinfo
Vm manager - qemu - new - import - https://download.cirros-cloud.net/0.5.1/

Exp6: file transfer
Auto-mount click
ls /media
ls /media/sf_VMShared
sudo usermod -aG vboxsf $USER
Sudo reboot
ls /media/sf_VMShared
VM1
cat /media/sf_VMShared/file.txt
touch vm1.txt
cp vm1.txt /media/sf_VMShared
VM2
cat /media/sf_VMShared/vm1.txt

nano /media/sf_VMShared/file.txt

cp ~/my_code.java /media/sf_VMShared/

cp /media/sf_VMShared/my_code.java ~/

Ubuntu to kali:
cp /home/osboxes/my_code.java /media/sf_VMShared/

cp /media/sf_VMShared/my_code.java /home/kali/

Kali to ubuntu:
cp /home/kali/notes.txt /media/sf_VMShared/

cp /media/sf_VMShared/notes.txt /home/osboxes/

Method 2:

Ip -4 a
Ip link

Sudo dhclient eth0
Ip -4 a
Sudo apt update
Sudo apt install nfs-kernel-server -y
Sudo mkdir -p /srv/nfs/nanfolder
Sudo chown nobody:nogroup /srv/nfs/nanfolder
Sudo chmod 777 /srv/nfs/nanfolder
Sudo nano /etc/exports
/srv/nfs/nanfolder <kali ip>(rw,sync,no_subtree_check,no_root_squash)
Sudo exportfs -a
Sudo systemctl restart nfs-kernel-server
Sudo ufw allow from <kali ip> to any port nfs
Kail:
Sudo wget https://archive.kali.org/archive-keyring.gpg -0 /usr/share/keyrings/kali-archive-keyring.gpg
Sudo apt update
Sudo apt install nfs-common -y
Sudo mkdir -p /mnt/nfs/nan
Sudo mout <ubuntu ip>:/srv/nfs/nanfolder /mnt/nfs/nanfolder
Df -h | grep nfs

Ubuntu:
Echo “Hello” | sudo tee /srv/nfs/nanfolder/ubuntu.txt
Kali:
Cat /mnt/nfs/nanfolder/ubuntu.txt

Exp7: hyper -v
Window:
Get-VMSwitch
New-Item -ItemType Directory -Path “C:\Hyper-V” -Force
New-VM -Name “Nan” -MemoryStartupBytes 2GB -Generation 2 -NewVHDPath “C:\Hyper-V\Nan.vhdx” -NewVHDSizeBytes 30GB
Connect-VMNetworkAdapter -VMName “Nan” -SwitchName “Default Switch”
Add-VMDvdDrive -VMName “Nan” -Path “ ubuntu image path”
Set-VMFirmware -VMName “Nan” -FirstBootDevice (Get-VMDvdDrive -VMName “Nan”)
Get-VM -Name “Nan”
Stop-VM -Name “Nan”
Get-VM -Name “Nan”

Exp8: open v switch

Ifconfig
Sudo apt-get install bridge-utils

```
Sudo modprobe bridge
Sudo modprobe br_netfilter
Sudo apt install openvswitch-switch -y
Sudo apt status openvswitch-switch

Sudo nano /ect/netplan/01-netcfg.yaml
Network:
        Version: 2
        Renderer: networkd
        Ethernets:
                Eno2:
                        Dhcp4: no
        bridges:
                Br-cloud:
                        Interfaces:[eno2]
                        Addresses:[]
                        Gateway4:
                        Nameservers:
                                Addresses:[8.8.8.8,1.1.1.1]
Sudo systemctl restart NetworkManager
Ip a
Route -n
Brctl show

Sudo brctl addbr br-cloud
Sudo brctl addif br-cloud-new enp0s3
Ifconfig enp0s3
Sudo Ifconfig br-cloud-new 10.0.2.15 netmask 255.255.255.0
Sudo Route add default gw 10.0.2.15 dev br-cloud-new

Sudo -s
Ip netns add VRF1
Ip netns add VRF2
Ip netns list

Ip link add veth1 type veth peer name veth2
Ip link show
Ip netns exec VRF1 ip link show
Ip link set veth1 netns VRF1
Ip link set veth3 netns VRF2
Ip netns exec VRF1 ip link
Ip netns exec VRF1 ifconfig veth1 10.10.10.1/24 up
Ip netns exec VRF2 ifconfig veth3 10.10.10.2/24 up
Ip netns exec VRF2 ifconfig
```

Ip netns exec VRF1 ifconfig

Sudo systemctl start openvswitch-switch
Sudo systemctl enable openvswitch-switch
Ovs-vsctl add-br vSwitch1
Ovs-vsctl show
Ovs-vsctl add-port vSwitch1 veth4
Ovs-vsctl set bridge vSwitch1 stp_enable=true
Ovs-appctl fdb/show vSwitch1

Exp9:
Sudo ip link set enp0s3
Sudo dhclient enp0s3
Ip a
Sudo apt update
Sudo apt upgrade -y
Sudo apt install -y git curl wget net-tools iproute2
Sudo adduser –shell /bin/bash/ –home /opt/stack stack
Echo “stack ALL=(ALL) NOPASSWD: ALL” | sudo tee /etc/sudoers.d/stack
Sudo chmod 0440 /etc/sudoers.d/stack
Sudo su - stack
Git clone https://git.openstack.org/opnstack/devstack
Cd devstack
Nano local.conf
Ip route get 1.1.1.1 | awk ‘{print $7}’ - not in /devstack

[[local|localrc]]
ADMIN_PASSWORD=dtuk2004
DATABASE_PASSWORD=$ADMIN_PASSWORD
RABBIT_PASSWORD=$ADMIN_PASSWORD
SERVICE_PASSWORD=$ADMIN_PASSWORD
DEST=/opt/stack
HOST_IP=10.0.2.15

Cd devstack
Export FORCE=yes
./stack.sh
http://localhost:8080/dashboard

Exp10:
Win+r
Diskmgmt.msc
Sudo -s
Fdisk -l

Pvcreate /dev/sdb and sdc
Vgcreate Nan /dev/sdb and Nana sdc
Pvdisplay
Vgdisplay
Lvcreate -l 100%FREE -n my_lv Nan
Lvdisplay
Vgextend Nan /dev/sdc
Lvdisplay
Lvresize - -size +3G /dev/Nan/my_lv
Df -h
Resize2fs /dev/sda1
Df -h